2025

Bitcoin Mining Economics

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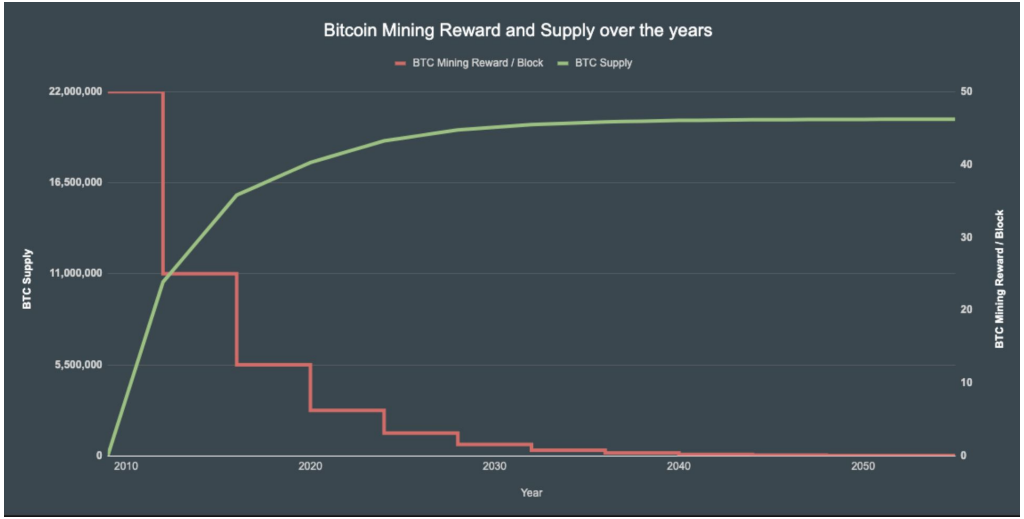
Agenda

1. Network Economics
2. Miner Economics
3. Mining Amortization Cycle
4. Competitivity is everything
5. Extra

Network Economics

Revenues: new emission subsidy + tx fees



Network Economics

The difficulty: how much work is needed to mine

Miner Economics

Costs

Capex:

- Data Center (Miners, Coolers, ecc)
- Electrical Infrastructure

Opex:

- Energy
- O&M

Miner Economics

Revenues - Hashprice

$$Hashprice = \frac{(Subsidy + Fees) * Bitcoin\ Price}{Difficulty * 2^{32}}$$

Miner Economics

Revenues - Energy Revenue

		Network Hashprice [$/PH/s/day]												
		40	50	60	70	80	90	100	110	120	130	140	150	160
Miner Efficiency [W/TH/s]	5	$333.33	$416.67	$500.00	$583.33	$666.67	$750.00	$833.33	$916.67	$1,000.00	$1,083.33	$1,166.67	$1,250.00	$1,333.33
	10	$166.67	$208.33	$250.00	$291.67	$333.33	$375.00	$416.67	$458.33	$500.00	$541.67	$583.33	$625.00	$666.67
	15	$111.11	$138.89	$166.67	$194.44	$222.22	$250.00	$277.78	$305.56	$333.33	$361.11	$388.89	$416.67	$444.44
	20	$83.33	$104.17	$125.00	$145.83	$166.67	$187.50	$208.33	$229.17	$250.00	$270.83	$291.67	$312.50	$333.33
	25	$66.67	$83.33	$100.00	$116.67	$133.33	$150.00	$166.67	$183.33	$200.00	$216.67	$233.33	$250.00	$266.67
	30	$55.56	$69.44	$83.33	$97.22	$111.11	$125.00	$138.89	$152.78	$166.67	$180.56	$194.44	$208.33	$222.22

Mining Amortization Cycle

Duration and amortization of a mining investment

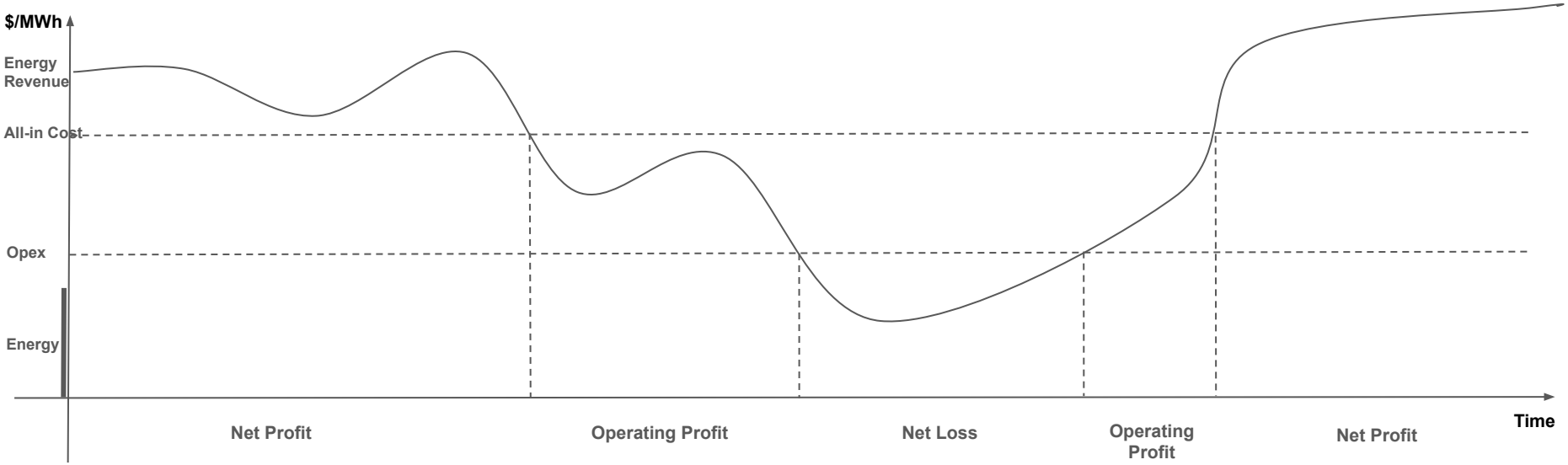
- **Duration: 3/4/5 years**

- **All-in cost: Capex [$/MWh] + Opex [$/Mwh]**
- **Revenues: (Hashprice [$/PH/s/day] / Efficiency [W/TH/s]) [$/MWh]**

- **Margin: Revenues - All-in cost**

Competitivity is Everything

The focus has to be on reducing costs



- Hashpower is a commodity, hashprice is defined by the market
- The only variables you can work on as a miner are efficiency and energy cost

Profitability of Bitcoin Mining

Bitcoin Mining is a **Cost** and **Risk Management** strategy: Having lower costs and good risk management practices makes you significantly less sensitive to market dynamics

$$Gross\ Profit = \left(\frac{Miner\ Hash\ Rate}{Network\ Hash\ Rate}\right) * (Price\ of\ BTC) * (BTC\ Daily\ Subsidy) - (Direct\ Operational\ Cost)$$

Alternative way

$$Gross\ Profit = (Hash\ Price - Hash\ Cost) * Miner\ Hash\ Rate$$

Can be controlled by Miner | **Cannot be controlled by Miner**

Term	Definition	Controlled by Miner	
		Yes	No
Miner Hash Rate	Hash Rate of Miner	x	
Network Hash Rate	Hash Rate of the Bitcoin Network		x
Price of BTC	Bitcoin Price		x
BTC Daily Subsidy	Bitcoin rewards to miners per day; includes the newly created BTC and the transactions fees.		x
Direct Operational Cost	Cost of Operation, includes Electricity cost, Direct Labor Costs and Regular Maintenance	x	

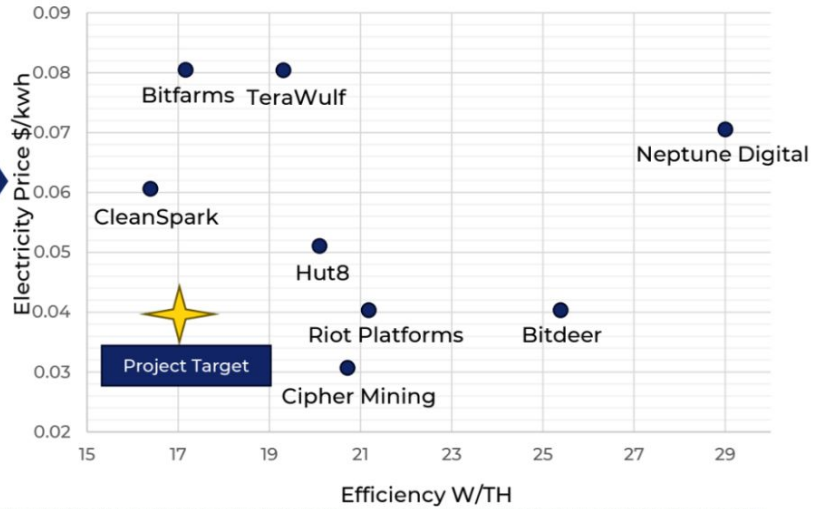
Hash Cost as a Measurement of Cost

Hash Cost is the Key Driver of Profitability

Given that revenue is calculated in terms of hash price, it's beneficial to express costs using the same metric, which facilitates a more straightforward assessment of a miner's profitability.

The total cost for operating a mining rig is primarily influenced by two variables: hashing efficiency and electricity cost. The hash cost is calculated using the formula below:

$$Hash\ Cost = (\ Electricity\ Cost\ for\ a\ Day) * (Hashing\ Efficiency)$$



By relentlessly managing our Cost Inputs and maximizing our Hashing Efficiency, we achieve a superior market position

$$Gross\ Profit = (Hash\ Price - Hash\ Cost) * Miner\ Hash\ Rate$$

Questions?

Thank you!

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